

Questions

❖ What are Total Sales and Profit Margin?

Total_Sales	Total_Quantity	Total_Profit	Profit_Margin_Percentage
2297200.86	37873.0	286397.02	12.47

- This is the Total Sales and Profit over the Companies dataset from 2014-2017.
- This data shows the company is profitable overall, but the profit margin is relatively low.
- This suggests that there is room to improve profitability, either by increasing revenue or reducing costs.

❖ Top 5 Products Sold by Quantity?

Product Name	Total_Quantity	Total_Profit	Profit_Margin_Percentage
Staples	215	292.2	38.68
Staple envelope	170	751.63	44.56
Easy-staple paper	150	1096.03	43.77
Staples in misc. colors	86	124.05	25.91
KI Adjustable-Height Table	74	-248.48	-5.46

- This table shows the most sold items during the companies' longevity of 2014-2017.
- Although Staples sold the most units, it was not the most profitable product.
- Staple envelopes generated higher profit despite selling fewer units.

❖ Top 5 Products by Revenue?

Product Name	Total_Revenue	Total_Profit	Total_Quantity	Profit_Margin_Percentage
Canon imageCLASS 2200 Advanced Copier	61599.82	25199.93	20	40.91
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.38	7753.04	31	28.24
Cisco TelePresence System EX90 Videoconferencing Unit	22638.48	-1811.08	6	-8.0
HON 5400 Series Task Chairs for Big and Tall	21870.58	0.0	39	0.0
GBC DocuBind TL300 Electric Binding System	19823.48	2233.51	37	11.27

- This data table shows the best products by revenue. The Canon imageCLASS 2200 Advanced Copier generates the highest profit by a significant margin.
- This product is highly valuable to the business due to its strong revenue and profit.

❖ Revenue by category?

Category	Total_Revenue	Total_Quantity	Total_Profit	Profit_Margin_Percentage
Technology	836154.03	6939.0	145454.95	17.4
Office Supplies	719047.03	22906.0	122490.8	17.04
Furniture	741999.8	8028.0	18451.27	2.49

- This table shows the three categories the company sells and compares their revenue and profit.
- Technology generates the highest revenue and profit, with Office Supplies also performing strongly.
- A key concern is the Furniture category, which has very low profitability. This may indicate the company is allocating resources to a low-performing category, where many products are either breaking even or generating losses.

❖ Revenue by subcategory?

Sub-Category	Total_Revenue	Total_Quantity	Total_Profit	Profit_Margin_Percentage
Phones	330007.05	3289.0	44515.73	13.49
Chairs	328449.1	2356.0	26590.17	8.1
Storage	223843.61	3158.0	21278.83	9.51
Tables	206965.53	1241.0	-17725.48	-8.56
Binders	203412.73	5974.0	30221.76	14.86

- The data above shows which sub-categories generated the highest revenue along with their profit margins.
- Tables stand out because they generate high revenue but result in a significant loss, suggesting a pricing issue or high costs.

❖ Revenue by state?

State	Total_Revenue	Total_Quantity	Total_Profit	Profit_Margin_Percentage
California	457687.63	7667.0	76381.39	16.69
New York	310876.27	4224.0	74038.55	23.82
Texas	170188.05	3724.0	-25729.36	-15.12
Washington	138641.27	1883.0	33402.65	24.09
Pennsylvania	116511.91	2153.0	-15559.96	-13.35

- The table above shows geographic revenue and profitability by state.
- Texas and Pennsylvania stand out as major loss-generating states.

❖ Revenue by city?

City	Total_Revenue	Total_Quantity	Total_Profit	Profit_Margin_Percentage
New York City	256368.16	3417.0	62036.98	24.2
Los Angeles	175851.34	2879.0	30440.76	17.31
Seattle	119540.74	1590.0	29156.1	24.39
San Francisco	112669.09	1935.0	17507.39	15.54
Philadelphia	109077.01	1981.0	-13837.77	-12.69

- The revenue by city data table shows the top five cities that generated the highest revenue.
- It also shows that though Philadelphia made top 5 highest revenue cities it still generated significant losses.

Queries

Total Sales

```
SELECT ROUND(SUM(Sales),2) AS Total_Sales, ROUND(SUM(Quantity),2)
AS Total_Quantity, ROUND(SUM(Profit),2) AS Total_Profit,
ROUND(SUM(Profit) / (SUM(Sales)) * 100,2) AS Profit_Margin_Percentage
FROM superstore;
```

Top 5 Products Sold by Quantity

```
SELECT "Product Name", SUM(Quantity) AS Total_Quantity
FROM Superstore
GROUP BY "Product Name"
ORDER BY Total_Quantity DESC
LIMIT 5;
```


Top 5 Products by Revenue

```
SELECT "Product Name", ROUND(SUM(Sales), 2) AS Total_Revenue,  
ROUND(SUM(Profit), 2) AS Total_Profit, SUM(Quantity) AS  
Total_Quantity, ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS  
Profit_Margin_Percentage  
FROM Superstore  
GROUP BY "Product Name"  
ORDER BY Total_Revenue DESC  
LIMIT 5;
```

Revenue by Category

```
SELECT Category, ROUND(SUM(Sales),2) AS Total_Revenue,  
ROUND(SUM(Quantity),2) AS Total_Quantity, ROUND(SUM(Profit),2) AS  
Total_Profit, ROUND(SUM(Profit) / SUM(Sales) * 100,2) AS  
Profit_Margin_Percentage  
FROM superstore  
GROUP BY Category  
ORDER BY Total_Revenue DESC;
```

Revenue by Subcategory

```
SELECT "Sub-Category", ROUND(SUM(Sales),2) AS Total_Revenue,  
ROUND(SUM(Quantity),2) AS Total_Quantity, ROUND(SUM(Profit),2) AS  
Total_Profit, ROUND(SUM(Profit) / SUM(Sales) * 100,2) AS  
Profit_Margin_Percentage  
FROM superstore  
GROUP BY "Sub-Category"  
ORDER BY Total_Revenue DESC  
LIMIT 5;
```

Revenue by State

```
SELECT State, ROUND(SUM(Sales),2) AS Total_Revenue,  
ROUND(SUM(Quantity),2) AS Total_Quantity, ROUND(SUM(Profit),2) AS  
Total_Profit, ROUND(SUM(Profit) / SUM(Sales) * 100, 2) AS  
Profit_Margin_Percentage  
FROM superstore  
GROUP BY State  
ORDER BY Total_Revenue DESC  
LIMIT 5;
```

Revenue by City

```
SELECT City, ROUND(SUM(Sales),2) AS Total_Revenue,  
ROUND(SUM(Quantity),2) AS Total_Quantity, ROUND(SUM(Profit),2) AS  
Total_Profit, ROUND(SUM(Profit) / SUM(Sales) * 100, 2) AS  
Profit_Margin_Percentage  
FROM superstore  
GROUP BY City  
ORDER BY Total_Revenue DESC  
LIMIT 5;
```